

SCFHS ➡ Saudi Commission For Health Specialties

الهيئة السعودية للتخصصات الصحية

- 1- Develops **standards** governing the practice of health professions and code of ethics.
- 2- Supervise and evaluate **training** programs for HCP
- 3- Set controls and standards for practice of HCP
- 4- Develops **medical** education programs
- 5- Supervise and approve the **results** of specialized exams
- 6- Issue certificates (diploma, fellowship.....etc.)
- 7- Evaluates and **approve** health certificates
- 8- Coordinates with associations, schools inside and outside the kingdom

MOH ➡ Ministry Of Health

★ **Medication errors** reported to MOH

★ MOH is responsible for **vaccinations**.

- 1- **Set laws** concerning health care, clinical practice
- 2- Supervise health affairs
- 3- Set strategic plans for hospitals
- 4- Regulate drug **distribution** and usage
- 5- Determine the official role of each HCP
- 6- Check quality of clinical institutions (**Inspection**, fellow up)
- 7- Responsible for psychological health
- 8- Responsible for drug information resources
- 9- Only source to issue data and analysis of diseases in KSA
- 10- **Inspection** of restaurants and food resources

MOI ➡ Ministry Of Interior

- 1- Responsible for **licensing** narcotic drugs.

MOE ➡ Ministry Of Education

- 1- Responsible for **equivalence** of the pharmacy certificates coming from external countries.

SFDA ➡ Saudi Food and Drug Administration

- ★ Adverse drug reactions reports uploaded to SFDA
- ★ Drug shortage is reported to SFDA
- ★ Misleading advertisement is reported to SFDA
- ★ Loss of narcotic log book is reported to SFDA
- ★ Chief pharmacist signs the form that is sent to SFDA

- 1- Drug registration and pricing
- 2- Receives complains about drug products
- 3- Inspection of pharmaceutical firms
- 4- Setting laws for importing pharmaceutical and cosmetics
- 5- Setting laws concerning medical equipment.
- 6- Determine list of companies for dealing with outside KSA
- 7- Approving drug formulary.

Responsible for medication regulation.

IRB ➡ institutional review board . approve research

8-

CBAHI ➡ Saudi Central Board for Accreditation of Health care Institutions

المركز السعودي الاعتماد المنشآت الصحية

Grant healthcare accreditation to all governmental and •

-private healthcare facilities

منح الاعتماد الصحي لجميع مرافق الرعاية الصحية

• Responsible for setting the quality and safety standards to ensure a better and safer healthcare.

مسؤول عن وضع معايير الجودة والسلامة لضمان رعاية صحية أفضل وأكثر أمن .

المادة الثالثة:

تقتصر ملكية الصيدلية أو منشأة بيع المستحضرات العشبية أو مركز الاستشارات الدوائية وتحليل المستحضرات الصيدلانية على السعوديين ويشترط لمنح الترخيص ما يأتي:

أ. أن يكون المالك أو أحد الشركاء صيدلياً مرخصاً له بمزاولة المهنة ويستثنى من ذلك المنشآت القائمة إلا في حالات الوفاة أو البيع أو التنازل وللوزير الاكتفاء بشرط الجنسية في المناطق النائية التي تحددها اللائحة.

ب. أن يكون مدير الصيدلية أو المنشأة أو المركز صيدلياً سعودياً متفرغاً مرخصاً له بمزاولة المهنة وللوزير الإعفاء من شرط الجنسية إذا لم يتوفر العدد الكافي من الصيادلة السعوديين.

ج. أن تتوفر في الصيدلية أو المنشأة أو المركز الشروط والمواصفات التي تحددها اللائحة.

اللائحة:

١/٣ يشترط لمنح الترخيص للصيدليات ومنشآت بيع المستحضرات العشبية في مدن الرياض، مكة المكرمة، جدة، المدينة المنورة، الخبر، الدمام فقط أن يكون المالك أو أحد الشركاء صيدلياً سعودياً مرخصاً له بمزاولة المهنة ، ويكتفى بشرط الجنسية فيما عدا ذلك من مدن ومحافظات.

Free samples ,,,exam

المادة الخامسة عشرة:

لا يجوز الإتجار في عينات المستحضرات الصيدلانية والعشبية المعدة للتعريف بالمنتج.

اللائحة:

- ١/١٥ يجب أن يطلع على عبوة عينات المستحضرات الصيدلانية والعشبية بخط واضح وباللغة العربية عبارة "عينة مجانية".
- ٢/١٥ يجب أن يقتصر توزيع العينات المجانية على تعريف الأطباء والصيدالة على الدواء ولا يجوز بيعها أو صرفها للمرضى ولا توزيعها على الجمهور، ولا يسمح بالاحتفاظ بها في الصيدليات.
- ٣/١٥ يمنع توزيع أو تداول العينات المجانية للأدوية المخدرة أو المؤثرات العقلية.
- ٤/١٥ يجب ألا تتجاوز العينات المجانية المسموح بتوزيعها نسبة ١% من الكمية المستوردة أو المصنعة محلياً من الدواء.
- ٥/١٥ يحظر على الصيدلي الحصول على أي منفعة مادية أو عينية من شركات أو وكلاء أو موزعين أو مستودعات الأدوية بقصد الترويج أو التسويق، ودفع المريض تجاه دواء، أو منتج طبي، أو خدمة محددة من ضمن خدمات الرعاية الصيدلانية.

Drug recall

⚡ Action taken by a company at any time to remove a defective drug product from the market

Others?

Thalidomide in pregnancy? Exam ?

Phocomelia

(((Kefauver Harris Amendment)))

The amendment was a response to the thalidomide

Kefauver Harris Amendment strengthened the U.S. Food and Drug Administration's control of experimentation on humans and changed the way new drugs are approved and regulated

E-Health in 2030 → ☞ Maximum utilization of health resources

Decrease DM ,,cancer ,,heart disease

Pharmacy and Therapeutics (P&T) committee

☞ Decides which drugs will appear on that entity's drug formulary

☞ P&T committee includes physicians, pharmacists and nurses

The pharmacy and therapeutics committee
❖ The pharmacy and therapeutics committee(PTC) is an advisory group of the medical staff and serves as the organizational line of communication between the medical staff and the pharmacy department. ❖ he committee is composed of physicians, the pharmacist and the other health professionals selected with guidance of the medical staff. ❖ This committee assists in the formulation of broad professional policies regarding the evaluation, selection, procurement, distribution, use, safety procedures and other matters relating to drugs use in the hospital.
Composition of PTC
❖ At least three physicians ❖ A pharmacist ❖ A representative of the nursing staff ❖ A hospital administrator or his/her designated person and ex-officio member of the committee ❖ The physician may be appointed as the " Chairman " of P.T.C. ❖ The pharmacist is designated as the " Secretary " of the committee.
Responsibility
❖ Ensure safety medication to patients ❖ The preparations of a hospital formulary, ❖ Publishing of a pharmacy educational bulletin, ❖ The establishment of automatic stop orders for dangerous drugs, ❖ The supervision of investigational use drugs, ❖ The development of a program for reporting and investigating adverse drug reactions and ❖ Assisting in the preparation of emergency kits or carts for medical emergencies.
Committee Agenda
❖ Minutes of the previous meeting. ❖ Review of a specified of the Formulary for up-dating and deletion of products. ❖ New drugs which have become commercially available. ❖ Investigational use of drugs currently in use in the hospital. ❖ Review of adverse drug reactions reported in the hospital since the last meeting. ❖ Drug safety in the hospital. ❖ Slow moving medication.
Role of PTC in drug safety
❖ Drug safety is the moral, legal and professional obligation of pharmacist . ❖ It includes responsibility from dispensing of drugs to drug administration . ❖ Following guidelines may sub serve the committee in ascertaining the adequate safety factor of hospital pharmacy. ❖ The hospital must employ a qualified, at least, a registered pharmacist with at least B. Pharm degree as 'Chief pharmacist' and the rest are may be at least diploma holders in pharmacist.

Functions of PTC
❖ Preparation of hospital drug formulary. ❖ Selection of manufacturer and supplier, mode of procurement. ❖ Addition of new drugs, deletion of old drugs. ❖ Drugs to be supplied in OPD. ❖ Policy formulation for pharmacy and monitoring ❖ Budget demand for pharmacy ❖ Developing Drug Information System ❖ Checking of pharmacy records and drug quality ❖ Maintenance of drug standard and quality control ❖ Disposal of Expiry Drugs
Committee's role in the adverse drug reaction program
❖ A consequence of recent advances in drug therapy is the proportionate increase in drug reactions. Reaction caused by blood and plasma products need to be reported unless a chemical agent other than the basic substance is responsible. ❖ An adverse Drug Reaction(ADR) Report Form, should be prepared by the PTC and made available on every nursing station. ❖ Every case of adverse drug reaction must be reported by the attending doctor to the clinical pharmacologist, if one is available, otherwise to the chairman of PTC. ❖ The completed ADR form of any having adverse drug reaction, will remove from the medical record and forward to the chairman or clinical pharmacologist, after discharge of the patient.

Medication errors are reported to:

- 1- P&T committee
- 2- Medication safety committee

If patient own medication (not ordered by physician, expired, not clean) → disposal of the drug and the patient should be informed about that action.

➡ High Alert Medications (HAM)

Electrolytes

Sterile water for injection

Dextrose ≥ 20%

NaCl > 0.9%

KCl injectable

Adrenergic agonist

Adrenergic antagonists

Anesthetic agents

Antiarrhythmics

Antiviral agents

Fibrinolytics

Antiplatelet

Anticoagulant

Chemotherapy

Hormonal therapy

Oxytocin

Electrolytes

Dialysis solutions

Insulin

notropic drugs

Epidural and intrathecal solutions

Sulfonyl urea

Liposomal forms of drugs

Neuromuscular blocking agents

Narcotics and controlled drug

Parenteral nutrition preparations

Total parenteral nutrition solution (TPN)

✦ Notes

Specific label or color :

Parenteral nutrition bags → Remove from refrigerator 0.5 hour before usage

- ✦ . IV anesthesia & Skeletal muscle relaxant → Stocked in ER, OR, ICU
- ✦ Dextrose 50% (D50W) → for dialysis patients for treating sudden hypoglycemia.

- **Consultant and specialist** are only allowed to prescribe HAM
- **Telephone orders not allowed in HAM** → Verbal orders are allowed in case of emergency only.
 - Must be double checked before administration.
- Limit access to HAM • All IV HAM are administered via IV pump.
- All IV infusions are standardized in mMole or mequ.
- Using "U" instead of units is not accepted for insulin and Heparin
- Concentrated electrolytes **should not be stocked in patient care area**

❖ **Narcotic, controlled and psychotropic drugs**

Narcotic drugs

Addicted drugs reduce pain and induce **euphoria**, lead to ↑ **tolerance and psychological dependence**.

✓ Narcotic drugs → **Red** Label

Controlled drugs

✓ Controlled drug → **Yellow** Label

Drug with **potential to be abused or addicted** which is held under strict control

Psychotropic drugs

Affecting **mind, emotions and behavior**

❖ Patient own **narcotic and controlled** medication shall be kept in the inpatient pharmacy in narcotics and controlled drugs cabinet **separately from the regular stock** medication → **not kept in patient cassette or trolley**.

❖ Narcotics and controlled drugs are stored behind steel door with double lock.

❖ **No one allowed to enter inside narcotic and controlled drugs except:**

- 1- Pharmacy staff
- 2- Anesthesia department
- 3- Assigned nursing staff.

- Strength and quantity of narcotics and controlled drug must be written clearly in words and figures.

- No strike-over, erasers or misspelling of drug name, strength or quantity.

- ID number should be written with physician name.

- Physician cannot prescribe narcotic or controlled drug to himself or family.

- Empty ampoule returns to the pharmacy with documents.

- Use the closest size ampoule for required dose.
- Unused drugs (tablets) must be returned to the pharmacy in charge.
- Replace used ampoules from pharmacy **within 3 days** from date of Rx

Prescription dispensing

➡ Narcotics and controlled:

- **Injections not dispensed for outpatient** or ambulatory patients to be used outside
- Tablets or patches are allowed to be given to outpatients.

➡ Psychotropic

- IV or tablets are allowed to be given in OPD

1- Psychiatrist	• Dispense as long as indicated.
2- Neurologist, Neurosurgeon	• Dispense Maximum of 3 Months
3- Non-Psychiatrist	• Dispense Maximum of 1 Month. (consultant and specialist)
4- ER department	• Dispense Maximum of 1 day.

- ✚ If drug is not available → original stamped Rx by the pharmacy department & copy given to the patient to be dispensed from other hospital or street pharmacy & Copy kept in patient File.

Prescription validity:: All prescriptions (RXs) must be dated at the time of writing and are valid for filling in the pharmacy as follows ↓

Rx from emergency room ER	• Narcotic drugs → valid for 1 day (24 hours) • Psychotropic and controlled drugs → Valid for 3 days
Rx from Outpatient department (OPD) /clinic	Valid for 7 days (1 week)

- ✚ Tablets (Lorazepam, Phenobarbital, Clonazepam, Tramadol, Diazepam, and Oxycodone) Rx is valid for 1 week.

❖ **Documentation:**

Rx must be documented for ↓

2 years	• Psychotropic and controlled
3 years	• Narcotics
10 years	• Narcotics log book

❖ Rx must be in 3 copies ↓

1	Patient file
2	Patient himself
3	Original in the pharmacy with empty ampoules

- ✚ Every case must be documented in a separate request form.
- ✚ Unused narcotics and controlled drugs must be destroyed/discarded within 6 months.
- ✚ **Empty morphine vials are discarded after 6 months**

❖ Storage and stability

→ Temperature:

Room temp.	15C – 25C	
Cool place	8 C – 15C	
Refrigerator	2 C – 8 C → 35:46 Fahrenheit	Vaccine, Insulin, Human plasma
Freeze	-10C – -25C	OPV stored at -20

- Insulin Flexpen → stable for 1 month after first use → kept at refrigerator.
- Human plasma unused → Discard immediately
- Temperature of the refrigerator recorded daily at log sheet.
- Large size refrigerator → stored at main pharmacy equipped with a temperature monitoring device connecting with maintenance department.

❖ Expired drugs

- ✚ Outdated (Expired) medication returned to drug store.

Occurrence/variance report (OVR)

- If there are any discrepancies → OVR must be completed before end of the shift & discrepancies must be resolved.
- UN resolved discrepancies → reported immediately to director of the pharmaceutical service.
- Broken or lost ampoule policy: OVR must be completed by nurse submitted to the TQM department, give the written report to the pharmacy with 2 witnesses

Disposal of ampoules:

- Flushing → 30 seconds of running water
- Placing in sharp containers (Injection only)

Safe dispense:

- In Patient: dispense medications in unit dose system for 24 hours only.
- Medication prescription by generic name except when brand name is acceptable or required
- Pharmacist can dispense generic equivalent drug.
- If there is no generic equivalent drug but therapeutic equivalent → should call the physician:

- If accepted → must write order for medication

- If not → the chief pharmacist asks for direct purchase.

Medication purchase

If drug is not available:

- ✳ Pharmacy department make request to MOH medical supply.
- ✳ If a drug is finished (out of stock) chief pharmacist make direct purchase → Purchase form → Finance department.

❖ Roles

- **Pharmacist:** Monthly inspection of pharmacy care areas to ensure proper patient safety and stock storage.
- **Nurse:** Weekly inspection of pharmacy care areas to ensure proper patient safety and stock storage.
- **Physician's department head must** approve every single case.
- Only **Consultant physicians** are allowed to Rx formulary drugs for unapproved indication (Off-Label) → Dispensed as Unit-dose

Drugs prescribed only by consultants

- a.* Amphotericin B
- b.* Ciprofloxacin
- c.* Caspofungin
- d.* Colomycin
- e.* Imipenem/meropenem
- f.* Voriconazole
- g.* Linezolid
- h.* Tigecycline

Drugs prescribed only by consultants and specialist

- High cost medications
- ✓ Botox, Mabthera, Humira, Keppra, Femara, Entecavir
 - High alert medication
 - Narcotics
- ✓ Morphine, Pethidine, Fentanyl
 - Central stores medication
- ✓ Hepatitis B, C drugs
- ✓ Multiple sclerosis drugs
- ✓ Peg-interferon, Pegasys, Ribavirin, Zeffix

❖ Types of medical orders

1- STAT order

- Administered immediately to the patient.
- Stated at the top of Rx → dispensed within 30 minutes
- Nurse shouldn't leave the pharmacy without STAT order
- Labeled (STAT)

2- Emergency order

- Stated at the top of the Rx → dispensed immediately

3- Now order

- Immediately dispensing

4- PRN orders

- Must include indication for use and duration

5- Routine orders

- Valid for 7 days unless short time period is specified

6- Telephone order

- Must be signed within 24 hours from giving the order.
- Accepted only in situations requires action facility care of the patient & patient condition doesn't requires physician presence → Verified by read-back by 2 licensed HCP workers receiving the order

Verbal orders

- In Emergency → verified by repeat back → signed by HCP when emergency case ends

قوانين وصفات العيادات الخارجية

يسمح بصرف أدوية الأمراض المزمنة لمدة تكفي تسعين يوم

أي ثلاثة أشهر

صلاحية صرف الوصفة سبعة أيام أي اسبوع واحد.

Drug Information Unit (DIU)

❖ DI pharmacist

- Receive phone calls from HCPs and give the answers verbally and written
- IF HCPs require about toxic or poisonous drug should contact:
 1. Poison control and DIC in Jeddah
 2. Drug information national center

❖ **The Five Rights of Medication Administration:**

1. Right Patient
2. Right Drug
3. Right Dose
4. Right Route
5. Right Time

❖ **4P of marketing:**

- 1- Product
- 2- Price
- 3- Place
- 4- Promotion

❖ Prescription dispensing:

- Dispense the medications for **chronic diseases** every **90 days** (3 months)
- The maximum allowed amount of **diazepam** to dispense is **30 days** (1 month)

★ Lookalike soundalike (LASA) medications

Medications that is visually similar in physical appearance or packaging and name of medications that have spelling similarities and/or similar phonetics.

Examples:

- 1- DOPamine VS DOBUTamine
- 2- HYDRALazine VS HYDROXYzine

How to avoid LASA?

- 1- Physical separation
- 2- Use tall MAN lettering → Capitalization of some letters to differentiate between 2 similar medications.
- 3- Use alert stickers

High alert =



Hazourds =



Control =



LASA =



Narcotics =



Expire =



QUESTIONS

- ❖ Humira patient asking you about his concerns that Humira causes cancer, what will you tell him
 - a. Humira doesn't cause cancer
 - b. Talk to him about the benefits vs risks
 - c. Yes, it may cause cancer
- ❖ Medical representative give you sample, what will you do?
 - a. Refuse
 - b. Accept
 - c. Tell pharmacy director
- ❖ Physician prescribes 10 mg of drug; the technician gave the pharmacist 100 mg by mistake and the wrong dose was given to the patient. After 3 days the patient came to the ER because of overdose. What should be done?
 - a. Tell the physician
 - b. Tell the pharmacy manager
 - c. Report to SFDA
 - d. Tell the patient

Credit of health institution is issued from which?

CBAHI central board for accreditation of health institution

Want to open pharmacy but no enough Saudi pharmacists to hire which is true?

MOH may make an exemption of nationality

Pharmacist saw misleading advertisement online what to do? f you are pharmacist and see in social media Advertising about medicine is misleading what will do?

SFDA

If stealing happen to who the pharmacist must report?

Pharmacy manager

Who is responsible from formulary

Pharmacy therapeutic committee

An employee perfected his job and the manger wants to something what will the employee receive from his manger

Reward

Original copy of the controlled medication given to :

Pharmacy

Room temperature 68 to 77°F (20 to 25°C)

Freezer temperature -10°C to -25°C

Refrigerator temperature 2°C to 8°C 14

storage of narcotic prescriptions according to Saudi national regulations ?

36 months

The responsibility of pharmacist in hospital is ?

deal with drug interaction

Pharmacy and therapeutic committee ? maintain of formulary of drug

Diazepam or oxycodone Prescription valid until ? 7 days

The maximum allowed amount of diazepam to dispense? 30 days

Concor and (another drug) are look alike what we recommend to use in a purpose of decreasing error : ?

Sticker blue

Vincristine fatal when given intrathecal what the process make safe from this error while dispense to nurse ?

dispense drug in piggyback

Patient take two eye drop in same time should separate? Separate five minutes

Patient receive prescription 17 December 2018, has 5 refill , came back in April 2019 asking for refill

Have one more refill

Example of intentional non-adherence ?

Patient does not take medicines because of fear of side effect

Which one is of 5 rights of drug administration?

Nonverbal communication which one?

Tone of voice Rate and volume of speech Facial expression Eye contact
Gestures/touch

Patient information how to deal with between practitioner ?

do not share it with anyone

Machine improve patient compliance with medication

E pill reminder

First thing before starting treatment for smoking patient?

Determination of patient to stop smoking

How can you make sure that the patients understand his medications ?

By asking the patient to repeat and explain the information given to him

Unit dose system in hospital ?

to decrease medication error

In 1997 , mean age of population was 50 year . In 2020 the mean age of population become 47 year , What does that mean ?

Increase mortality

DR:KABSHA